Time Series Regression and Forecasting

Juergen Meinecke

Research School of Economics, Australian National University

Roadmap

Announcements

Time Series Terminology, Autocorrelation

Autoregressive Models and Forecasting

Week 9 Quiz: Canvas outage

There was a Canvas outage at the end of last week

Canvas was definitely not working Friday morning and early afternoon

But some students reported technical problems already on Thursday afternoon while working on their quizzes

Once Canvas was up and running again, I re-opened the quiz and extended the 'deadline' to Monday afternoon (to give students who were affected by the outage a full working day to work on the quiz)

Assignment 2

Due next week Wednesday at 5.00pm **at the latest**!

This deadline is sharp, no extensions! Really!

Covers course material up to (and including) week 9

My strong recommendation:
Submit your solution this week already!

Get it out of the way early!

Roadmap

Announcements

Time Series Terminology, Autocorrelation

- Lags, First Differences, Growth Rates
- Autocorrelation

Autoregressive Models and Forecasting

So far we have worked with **cross-sectional** data:
many observations at a single point in time
(e.g. earnings data for 17,870 workers in 1994 in our earnings-height or earnings-weight estimations)

Now we work with **time series** data: observations on a single entity recorded at successive points in time
(e.g. quarterly GDP for Australia from 1960 to 2025)

The fundamental new feature: **observations have a natural ordering**

- In cross-sectional data, reshuffling the rows changes nothing
- In time series data, the past can influence the future — but not vice versa

This ordering creates both a **challenge** (observations are no longer independent — yesterday's inflation is correlated with today's) and an **opportunity** (we can use past values to forecast future values)

Notation is now slightly different

Instead of an i-subscript, variables will have a t-subscript
(this is not a substantive change, just convention for time series)

The variable Y_t is the value of Y (for example real GDP) in period t
(for example year)

Data set: $\{Y_1, \ldots, Y_T\}$ are T observations on the time series Y

We consider only consecutive, evenly-spaced observations
(for example, monthly, 1960 to 1999, no missing months)

Missing and unevenly spaced data do not pose a principal problem and only introduce technical complications which we are happy to ignore at this stage

Definition

The **first lag** of time series Y_t is Y_{t-1}.

The j**-th lag** of time series Y_t is Y_{t-j}.

Definition

The **first difference** of time series Y_t is $\Delta Y_t := Y_t - Y_{t-1}$.

Definition

The **first difference of the logarithm** of time series Y_t is $\Delta \ln(Y_t) := \ln(Y_t) - \ln(Y_{t-1})$.

With these definitions it is easy to determine the percentage change of a time series Y_t between the periods $t-1$ and t: it is approximately $100 \cdot \Delta \ln(Y_t)$

Example: Quarterly CPI data for the US

I'm starting out with a time series on the price level in the US

Price level here is measured by the consumer price index (CPI)

The specific time series I'm using is labelled `CPIAUCSL`

It is the *Consumer Price Index for All Urban Consumers* provided by the Federal Reserve Bank of St. Louis (FRED)

Let's look at two recent measurements

- CPI in the fourth quarter of 2025 (2025:Q4) = 325.55
- CPI in the first quarter of 2026 (2026:Q1) = 328.11

Given this price level data, how do we back out inflation?

We study two approaches: exact and approximate

- CPI in the fourth quarter of 2025 (2025:Q4) = 325.55
- CPI in the first quarter of 2026 (2026:Q1) = 328.11
- Inflation via *exact* percentage change in CPI, 2025:Q4 to 2026:Q1

$$100 \cdot \left(\frac{328.11 - 325.55}{325.55} \right) = 0.789\%$$

- Inflation via logarithmic *approximation* instead:

$$100 \cdot (\ln(328.11) - \ln(325.55)) = 0.785\%$$

The two approaches give slightly different results

It is common to extrapolate up the quarter-to-quarter change to an annual rate

Quarter-to-quarter change *at an annual rate*

- Annualized inflation via *exact* percentage change in CPI

$$4 \cdot 100 \cdot \left(\frac{328.11 - 325.55}{325.55} \right) = 3.154\%$$

- Annualized inflation via logarithmic *approximation* instead:

$$4 \cdot 100 \cdot (\ln(328.11) - \ln(325.55)) = 3.142\%$$

Answers the question: if the current quarter inflation continued throughout the year, what would annual inflation be?

It’s a simple extrapolation really

Roadmap

Announcements

Time Series Terminology, Autocorrelation

- Lags, First Differences, Growth Rates
- Autocorrelation

Autoregressive Models and Forecasting

The correlation of a time series with its own lagged values is called autocorrelation or serial correlation

Why does autocorrelation arise? Many economic variables exhibit *persistence*: high inflation today tends to be followed by high inflation tomorrow; a recession this quarter makes a recession next quarter more likely. Autocorrelation is simply the statistical signature of this persistence.

Definition

The j-th **autocovariance** of a time series Y_t is the covariance between Y_t and its j-th lag, Y_{t-j}: $\text{Cov}(Y_t, Y_{t-j})$.

The j-th **autocorrelation** of a time series Y_t is the correlation between Y_t and its j-th lag, Y_{t-j}:

$$\rho(j) := \frac{\text{Cov}(Y_t, Y_{t-j})}{\sqrt{\text{Var}\,(Y_t)\text{Var}\,(Y_{t-j})}}.$$

The sample autocorrelation is the *estimated* autocorrelation

Definition

The j-th **sample autocorrelation** of a time series Y_t is the correlation between Y_t and its j-th lag, Y_{t-j}:

$$\hat{\rho}(j) := \frac{\widehat{\text{Cov}}(Y_t, Y_{t-j})}{\widehat{\text{Var}}\,(Y_t)},$$

$$\text{with} \quad \widehat{\text{Cov}}(Y_t, Y_{t-j}) := \frac{1}{T}\sum_{t=j+1}^{T}(Y_t - \overline{Y}_{j+1,T})(Y_{t-j} - \overline{Y}_{1,T-j})$$

$$\overline{Y}_{p,q} := \frac{1}{T-j}\sum_{t=p}^{q} Y_t$$

$$\widehat{\text{Var}}\,(Y_t) := \frac{1}{T}\sum_{t=1}^{T}(Y_t - \overline{Y})^2$$

Two little comments:

Although we only compare $T - j$ pairs of the time series, the division is by T (this is conventional in time series analysis)

When computing the sample autocorrelation, we have implicitly assumed that

- variances are constant over time
- covariances are constant over time
  (only dependent on the lag length j)

This is justified by *stationarity* (which we will define next week)

Python example: quarterly CPI data for the US

Using the time series CPIAUCSL on quarterly CPI in the US,
I create the quarter-to-quarter inflation at an annualized rate

Python Code

```
> import pandas as pd
> import statsmodels.formula.api as smf
> import numpy as np
> # reading data from spreadsheet (downloaded from FRED):
> df = pd.read_csv('CPIAUCSL.csv')

> # creating quarterly index
> df['date'] = pd.to_datetime(df['DATE'], format='%Y-%m-%d')
> df.index = pd.DatetimeIndex(df.date, name='quarter').to_period('Q')

> # copy of CPI series with easy-to-access name:
> df['cpi'] = df.CPIAUCSL

> # taking logarithm of original series:
> df['logcpi'] = np.log(df.cpi)

> # creating annualised inflation via differences in logs:
> # (this is the 'first derivative' of 'cpi')
> df['infl'] = 400 * df.logcpi.diff()

> # creating quarter-on-quarter differences in inflation:
> # (this is the 'second derivative' of 'cpi')
> df['dinfl'] = df.infl.diff()

> df = df.drop(['DATE', 'CPIAUCSL'], axis=1)
```

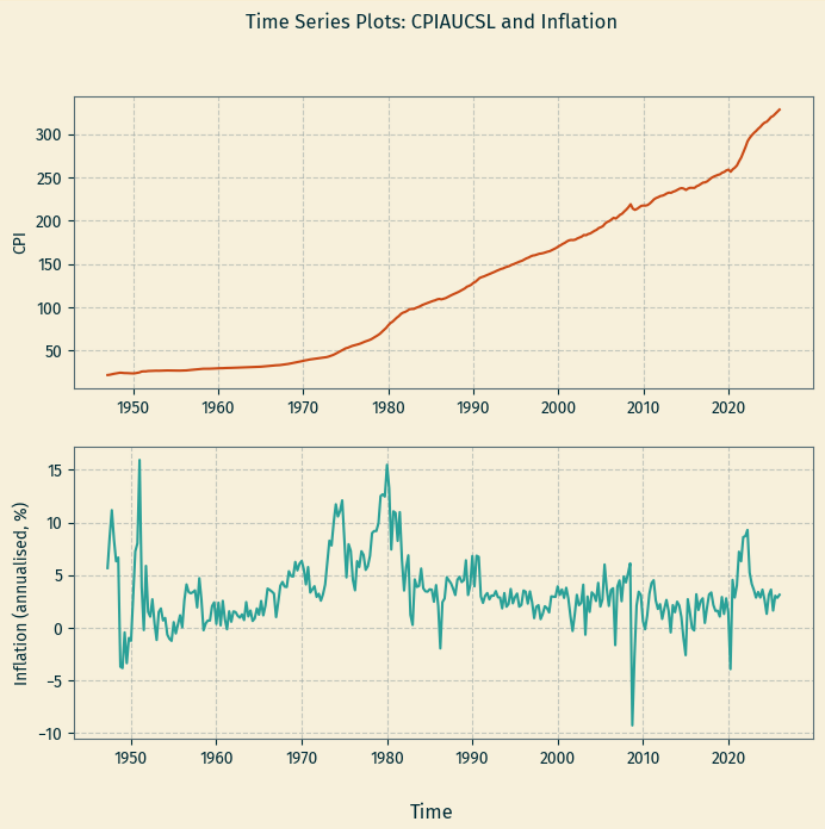
Let’s take a look at the time series

Python Code

```
> # looking at data: top two years
> print(df.head(8))
             date     cpi    logcpi       infl      dinfl
quarter
1947Q1  1947-01-01  21.700  3.077312        NaN        NaN
1947Q2  1947-04-01  22.010  3.091497   5.673854        NaN
1947Q3  1947-07-01  22.490  3.113071   8.629548   2.955694
1947Q4  1947-10-01  23.127  3.141001  11.172000   2.542452
1948Q1  1948-01-01  23.617  3.161967   8.386410  -2.785590
1948Q2  1948-04-01  23.993  3.177762   6.318132  -2.068278
1948Q3  1948-07-01  24.397  3.194460   6.679221   0.361089
1948Q4  1948-10-01  24.173  3.185236  -3.689546 -10.368768

> # looking at data: bottom two years
> print(df.tail(8))

             date      cpi    logcpi      infl     dinfl
quarter
2024Q2  2024-04-01  313.081  5.746462  2.663754 -0.964190
2024Q3  2024-07-01  314.121  5.749778  1.326528 -1.337226
2024Q4  2024-10-01  316.588  5.757601  3.129193  1.802665
2025Q1  2025-01-01  319.475  5.766679  3.631112  0.501919
2025Q2  2025-04-01  320.786  5.770774  1.638084 -1.993028
2025Q3  2025-07-01  323.235  5.778380  3.042151  1.404067
2025Q4  2025-10-01  325.547  5.785507  2.850893 -0.191259
2026Q1  2026-01-01  328.114  5.793361  3.141706  0.290813
```

Python Code

```
> fig, axs = plt.subplots(2, 1, figsize=(8,7))
> axs[0].plot(df.date, df.cpi)
> axs[0].set_ylabel('CPI')
> axs[1].plot(df.date, df.infl)
> axs[1].set_ylabel('Inflation (annualised, %)')
> fig.supxlabel('Time')
> fig.suptitle('Time Series Plots: CPIAUCSL and Inflation')
> plt.show()
```



Then I look at sample autocorrelations

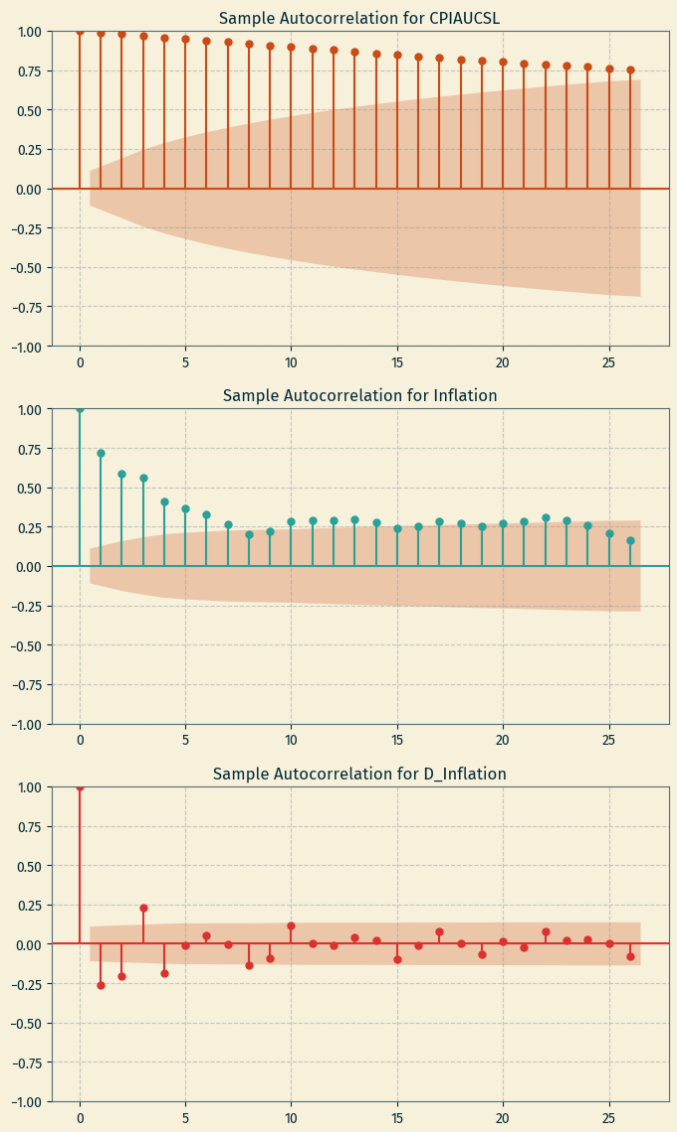
Python Code

```
> from matplotlib import pyplot as plt
> from statsmodels.graphics.tsaplots import plot_acf

> # creating a 'stacked' plot of 3 rows
> fig, axs = plt.subplots(3, 1, figsize = (8, 14))

> # stacking them
> plot_acf(df.cpi, ax=axs[0], title = 'Sample Autocorrelation for CPIAUCSL')
> plot_acf(df.infl, missing='drop', ax=axs[1], title = 'Sample Autocorrelation for Inflation')
> plot_acf(df.dinfl, missing='drop', ax=axs[2], title = 'Sample Autocorrelation for D_Inflation')
> plt.show()
```

which creates the following plot ...

Increasing degree of ‘differentiation’ reduces autocorrelation

Python Code Output



These sample autocorrelations show

- the original time series `CPIAUCSL` (price level as measured by cpi) is very highly serially or auto-correlated
- `infl` (the first derivative of `CPIAUCSL`) is still highly serially correlated
- `dinfl` (the first derivative of `infl` and second derivative of `CPIAUCSL`) is not serially correlated anymore

Please bear this in mind, as it will have important ramifications when we want to run auto-regressions using price level or inflation data

Detecting serial correlation by visual inspection is tricky:
The `infl` series looks “noisy” yet has strong autocorrelation
Always use the autocorrelation plot, not just the time series plot

From Autocorrelation to Forecasting

The autocorrelation plots reveal something economically meaningful:
past values of a time series are informative about its future values

This suggests a natural forecasting strategy:
regress Y_t on its own past values $Y_{t-1}, Y_{t-2}, \ldots$

This class of models is called **autoregressions** (AR models)

They are the workhorse of time series forecasting

Roadmap

Announcements

Time Series Terminology, Autocorrelation

Autoregressive Models and Forecasting

- The First Order Autoregressive (AR(1)) Model
- The p-th Order Autoregressive (AR(p)) Model

A natural starting point for a forecasting model is to use past values of Y (that is, $Y_{t-1}, Y_{t-2}, \dots$) to forecast Y_t

An autoregression is a regression model in which Y_t is regressed against its own lagged values

The number of lags used as regressors is called the *order* of the autoregression

In a first order autoregression, Y_t is regressed against Y_{t-1}

In a p-th order autoregression, Y_t is regressed against $Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}$

The population AR(1) model is

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$$

The coefficient β_1 does *not* have a causal interpretation. We are not asking "what happens to Y_t if we could force Y_{t-1} to change?"

We are simply using the historical relationship between Y_t and Y_{t-1} to predict.

If $\beta_1 = 0$ then Y_{t-1} carries no information about Y_t;
past values are useless for forecasting

The AR(1) model is estimated by OLS regression of Y_t on Y_{t-1}

Testing $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$ is a test of whether Y_{t-1} has any predictive content for Y_t

Python Code

```
> # creating lagged inflation
> # (will be used as explanatory variable in AR(1) estimation)
> df['l1infl'] = df.infl.shift(1)

> # looking at data: top two years
> print(df[['cpi', 'infl', 'l1infl']].head(8))
            cpi       infl     l1infl
quarter
1947Q1   21.700        NaN        NaN
1947Q2   22.010   5.673854        NaN
1947Q3   22.490   8.629548   5.673854
1947Q4   23.127  11.172000   8.629548
1948Q1   23.617   8.386410  11.172000
1948Q2   23.993   6.318132   8.386410
1948Q3   24.397   6.679221   6.318132
1948Q4   24.173  -3.689546   6.679221


> # looking at data: bottom two years
> print(df[['cpi', 'infl', 'l1infl']].tail(8))
             cpi      infl    l1infl
quarter
2024Q2   313.081  2.663754  3.627944
2024Q3   314.121  1.326528  2.663754
2024Q4   316.588  3.129193  1.326528
2025Q1   319.475  3.631112  3.129193
2025Q2   320.786  1.638084  3.631112
2025Q3   323.235  3.042151  1.638084
2025Q4   325.547  2.850893  3.042151
2026Q1   328.114  3.141706  2.850893
```

Here I'm running an AR(1) estimation for `infl`

Python Code (output edited)

```
> # first order autoregression:
> ar1 = smf.ols('infl ~ l1infl', data=df, missing='drop').fit(use_t=False)
> print(ar1.summary())

OLS Regression Results
==============================================================================
Dep. Variable:                   infl   R-squared:                       0.521
Model:                            OLS   Adj. R-squared:                  0.520
Method:                 Least Squares   F-statistic:                     340.6
No. Observations:                 315
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.9503      0.182      5.213      0.000       0.593       1.308
l1infl         0.7213      0.039     18.456      0.000       0.645       0.798
==============================================================================
```

Notice: We don't need to use heteroskedasticity-robust standard errors because we are not really interested in statistical inference, instead we want to use the coefficient estimates to produce forecasts

Forecasting

Our main objective when estimating autoregressions is to produce *forecasts*

We are not interested in causal effects

As a consequence, we are not usually interested in the coefficient estimates of AR models

We only use the coefficient estimates to create a forecast for the dependent variable

External validity is paramount: the model estimated using historical data must hold into the (near) future

But what do I mean by *forecast*?

Notation

- For an AR(1) model:

$$Y_{T+1|T} = \beta_0 + \beta_1 Y_T$$

$$\hat{Y}_{T+1|T} = \hat{\beta}_0 + \hat{\beta}_1 Y_T$$

- $Y_{T+1|T}$: forecast of Y_{T+1} based on $Y_T, Y_{T-1}, \ldots$ using the population coefficients (typically unknown)
- $\hat{Y}_{T+1|T}$: forecast of Y_{T+1} based on $Y_T, Y_{T-1}, \ldots$ using the estimated coefficients
- Forecast errors are defined by $Y_{T+1} - \hat{Y}_{T+1|T}$

Do not confuse predicted values with forecasts

- *Predicted values* are "in-sample"
- *Forecasts* are "out-of-sample" (looking into the future)

Let me explain the difference between predicted values and forecasts

Earlier we estimated the following AR(1) model for inflation:

$$\widehat{\texttt{infl}}_t = 0.9503 + 0.7213 \cdot \texttt{infl}_{t-1}$$

We used data from 1947:Q1–2026:Q1 for the estimation

This means:

- $\widehat{\texttt{infl}}_{2026:Q1}$ is a predicted value
- $\widehat{\texttt{infl}}_{2026:Q2|2026:Q1}$ is a forecast

Let's calculate both

These are simple common sense calculations

Calculation for the predicted value

In the data we observe $\texttt{infl}_{2025:Q4} = 2.8509$

Resulting in the predicted value

$$\widehat{\texttt{infl}}_{2026:Q1} = 0.9503 + 0.7213 \cdot 2.8509 = 3.0067$$

In my data set I do observe $\texttt{infl}_{2026:Q1} = 3.1417$
therefore $\texttt{infl}_{2026:Q1} - \widehat{\texttt{infl}}_{2026:Q1}$ is the *residual* for Q1 2026

Calculation for the forecast

In the data we observe $\texttt{infl}_{2026:Q1} = 3.1417$

Resulting in the forecast values

$$\widehat{\texttt{infl}}_{2026:Q2|2026:Q1} = 0.9503 + 0.7213 \cdot 3.1417 = 3.2165$$

I could wait until July when $\texttt{infl}_{2026:Q2}$ is released and calculate the *forecast error* $\texttt{infl}_{2026:Q2} - \widehat{\texttt{infl}}_{2026:Q2|2026:Q1}$

Easy to produce predicted values and forecasts in `Python`

Just use the post-regression `predict` function

It will produce a predicted value when in-sample

It will produce a forecast value when out-of-sample

Python Code

```
> # Prediction for 2026:Q1, and forecast for 2026:Q2
> newdata = {'l1infl' : [df.infl[-2], df.infl[-1]]}
> ar1.predict(newdata)

0    3.006735
1    3.216508
```

Roadmap

Announcements

Time Series Terminology, Autocorrelation

Autoregressive Models and Forecasting

- The First Order Autoregressive (AR(1)) Model
- The p-th Order Autoregressive (AR(p)) Model

Why More Than One Lag?

The AR(1) model is a natural starting point, but one lag may not be enough:

- Seasonal patterns: quarterly inflation may have a quarterly seasonal cycle, so inflation four quarters ago (Y_{t-4}) may matter as much as last quarter
- Sluggish adjustment: shocks to the economy may take several periods to work through, requiring multiple lags to capture the dynamics
- Remaining autocorrelation: if the AR(1) residuals are still serially correlated, adding more lags can absorb that autocorrelation and improve forecasts

The general AR(p) model addresses these concerns by including the p most recent lags

The population AR(p) model is

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \cdots + \beta_p Y_{t-p} + u_t$$

Again, the coefficients do NOT have a causal interpretation

Testing whether extra lags help: to test whether $Y_{t-2}, \ldots, Y_{t-p}$ add predictive value over and above Y_{t-1}, use an F-test of $H_0 : \beta_2 = \cdots = \beta_p = 0$

Choosing optimal p:

- an appropriate lag length might be implied by an economic model; for example central banks employ time series models with a lag of at least 6 to 8 to be able to capture the transmission mechanism of monetary policy
- alternatively, there are purely data driven statistical information criteria (called BIC and AIC) that trade off model precision versus complexity

Here I'm preparing an AR(4) estimation for `infl`

Python Code

```
> # creating more lags for inflation
> df['l2infl'] = df.infl.shift(2)
> df['l3infl'] = df.infl.shift(3)
> df['l4infl'] = df.infl.shift(4)

>#  looking at data: top two years
> print(df[['cpi', 'infl', 'l1infl', 'l2infl', 'l3infl', 'l4infl']].head(8))
            cpi       infl     l1infl     l2infl     l3infl     l4infl
quarter
1947Q1   21.700        NaN        NaN        NaN        NaN        NaN
1947Q2   22.010   5.673854        NaN        NaN        NaN        NaN
1947Q3   22.490   8.629548   5.673854        NaN        NaN        NaN
1947Q4   23.127  11.172000   8.629548   5.673854        NaN        NaN
1948Q1   23.617   8.386410  11.172000   8.629548   5.673854        NaN
1948Q2   23.993   6.318132   8.386410  11.172000   8.629548   5.673854
1948Q3   24.397   6.679221   6.318132   8.386410  11.172000   8.629548
1948Q4   24.173  -3.689546   6.679221   6.318132   8.386410  11.172000

> # looking at data: bottom two years
> print(df[['cpi', 'infl', 'l1infl', 'l2infl', 'l3infl', 'l4infl']].tail(8))
             cpi      infl    l1infl    l2infl    l3infl    l4infl
quarter
2024Q2   313.081  2.663754  3.627944  2.873418  3.401471  2.868553
2024Q3   314.121  1.326528  2.663754  3.627944  2.873418  3.401471
2024Q4   316.588  3.129193  1.326528  2.663754  3.627944  2.873418
2025Q1   319.475  3.631112  3.129193  1.326528  2.663754  3.627944
2025Q2   320.786  1.638084  3.631112  3.129193  1.326528  2.663754
2025Q3   323.235  3.042151  1.638084  3.631112  3.129193  1.326528
2025Q4   325.547  2.850893  3.042151  1.638084  3.631112  3.129193
2026Q1   328.114  3.141706  2.850893  3.042151  1.638084  3.631112
```

Here I'm running an AR(4) estimation for `infl`

Python Code (output edited)

```
> # fourth order autoregression:
> ar4 = smf.ols('infl ~ l1infl + l2infl + l3infl + l4infl',
                data=df, missing='drop').fit(use_t=False)
> print(ar4.summary())

OLS Regression Results
==============================================================================
Dep. Variable:                   infl   R-squared:                       0.558
Model:                            OLS   Adj. R-squared:                  0.553
Method:                 Least Squares   F-statistic:                     97.03
No. Observations:                 312
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.7531      0.190      3.956      0.000       0.380       1.126
l1infl         0.6158      0.056     10.945      0.000       0.506       0.726
l2infl         0.0094      0.064      0.149      0.882      -0.115       0.134
l3infl         0.3130      0.064      4.925      0.000       0.188       0.438
l4infl        -0.1667      0.056     -2.995      0.003      -0.276      -0.058
==============================================================================
```

Again producing prediction and forecast

We estimated the following AR(4) model for inflation:

$$\widehat{\texttt{infl}}_t = 0.7531 + 0.6158 \cdot \texttt{infl}_{t-1} + 0.0094 \cdot \texttt{infl}_{t-2} + 0.3130 \cdot \texttt{infl}_{t-3} - 0.1667 \cdot \texttt{infl}_{t-4}$$

In the data we observe

Python Code

```
> df.infl.tail(5)
quarter
2025Q1    3.631112
2025Q2    1.638084
2025Q3    3.042151
2025Q4    2.850893
2026Q1    3.141706
```

$$\widehat{\texttt{infl}}_{2026:Q1} = 0.7531 + 0.6158 \cdot (2.8509) + 0.0094 \cdot (3.0422) + 0.3130 \cdot (1.6381) - 0.1667 \cdot (3.6311) = \mathbf{2.4448}$$

$$\widehat{\texttt{infl}}_{2026:Q2|2026:Q1} = 0.7531 + 0.6158 \cdot (3.1417) + 0.0094 \cdot (2.8509) + 0.3130 \cdot (3.0422) - 0.1667 \cdot (1.6381) = \mathbf{3.3939}$$

Forecasting

Still easy to produce predicted values and forecasts in Python

Again use the post-regression `predict` function

It will produce a predicted value when in-sample

It will produce a forecast value when out-of-sample

Python Code

```
> # Prediction for 2026:Q1, and forecast for 2026:Q2
> newdata = {'l1infl' : [df.infl[-2], df.infl[-1]],
             'l2infl' : [df.infl[-3], df.infl[-2]],
             'l3infl' : [df.infl[-4], df.infl[-3]],
             'l4infl' : [df.infl[-5], df.infl[-4]]}
> ar4.predict(newdata)

0    2.444769
1    3.393888
```